# Resonant cascaded Kerr nonlinearity in $\chi^{(2)}$ metasurfaces

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**Abstract:** A detuned second-harmonic resonance can replace the phase mismatch that sets the cascaded Kerr nonlinearity of bulk crystals. We derive the cascaded susceptibility of a general resonator, with an overlap bounded by one and the bulk result as a limit. Its sign follows the detuning $\Delta\omega$, it carries a nonlinear loss, and its product with the bandwidth does not depend on the quality factor. In doubly resonant $LiNbO_3$ metasurfaces with a second-harmonic quality factor of 4000 it is about 20 times the native Kerr response, and in poled microrings 400 to 3000 times. For a singly resonant metasurface we propose a picosecond Z-scan that isolates the part of the response odd in $\Delta\omega$.

## 1. Introduction

1. Kerr nonlinearity underlies all-optical switching, optical bistability, parametric oscillation and frequency combs. The native $\chi^{(3)}$ of low-loss dielectrics is small, about $3\times10^{-21}$ m$^2$/V$^2$ in $LiNbO_3$ (Supplement 1, Section 5), and its sign is set by the material. Field enhancement in resonant nanostructures increases the nonlinear phase per unit power without changing either [1].

2. Two cascaded second-order processes act on the pump as a Kerr medium. The effect was predicted by Ostrovskii [2], observed in KTP as a nonlinear refraction that changes sign with the phase mismatch [3], reviewed in Ref. [4] and described microscopically as a collective effect [5]. In bulk crystals its strength scales as $1/\Delta k$, with $\Delta k$ the phase mismatch, and it is used for soliton compression [6–8]. Khurgin and co-workers proposed a second-harmonic resonance in place of $\Delta k$ for surface-emitting structures [9, 10], with an enhancement over the native response of the order of the quality factor of that resonance [5].

3. In $\chi^{(2)}$ cavities the eliminated second harmonic acts on the pump as a Kerr term with a nonlinear loss [11–14]. In a $LiNbO_3$ microdisk and in AlN microrings this term produced four-wave mixing [15, 16], and in an AlN microring the detuning of the second-harmonic mode tuned the self-phase modulation from −2.7 to +4.7 times its native value [17]. In free-space cavities around periodically poled $LiNbO_3$ crystals, an effective Kerr response estimated at $10^2$ to above $10^3$ times the native one drives dissipative quadratic solitons [18–20]. Four-wave mixing in a poled thin-film waveguide corresponds to an effective Kerr index $1.6\times10^4$ times the native value [21]. In nanoresonators and metasurfaces, quasi-bound states in the continuum enhance second-harmonic generation [22–26], also in $LiNbO_3$ [27], and cascaded second-order processes have produced third-harmonic light [28, 29]. The back-conversion of the second harmonic in a metasurface has been treated as pump depletion [30]; to our knowledge, the Kerr response it produces on the pump has not been evaluated or compared with the native response of the host.

4. We extend the resonant replacement of $\Delta k$ to a general resonator with vector modes. The

cascaded susceptibility takes one closed form with a geometric overlap $\eta_{\mathrm{geom}} \leq 1$. It scales as $\omega_1/\Delta\omega$, that is as the quality factor $Q_2$ of the second-harmonic mode at a fixed ratio of $\Delta\omega$ to its linewidth, and changes sign with $\Delta\omega$ [Fig. 1(a)]. For any resonator, the product of strength and bandwidth depends on neither $Q_2$ nor $\Delta\omega$, and the ratio $R$ of the cascaded to the native Kerr response of the same mode depends on the host through a single figure of merit. In $LiNbO_3$, $R$ is about 20 in a doubly resonant metasurface with $Q_2 = 4000$ and $\eta_{\mathrm{geom}} = 0.3$, with a bandwidth of 0.1 THz, and 400 to 3000 in poled microrings, with bandwidths below 1 GHz. In a microring the cascaded shift follows from the measured second-harmonic efficiency, so the formula can be tested with no free parameter. For a singly resonant metasurface, where the native response does not vary across the second-harmonic resonance, we propose a picosecond Z-scan that isolates the part of the response odd in $\Delta\omega$.

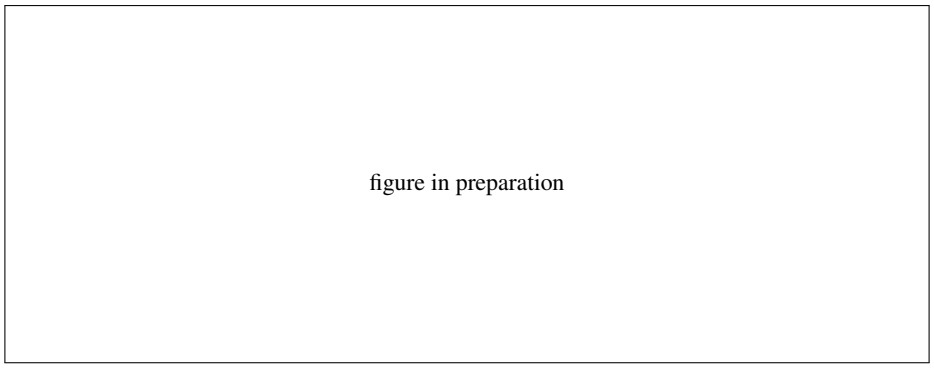



Fig. 1. Cascaded Kerr response at a second-harmonic resonance. (a) Second-harmonic generation and difference-frequency generation through a mode at $\omega_2$, detuned by $\Delta\omega = 2\omega_1 - \omega_2$. (b) Doubly resonant $LiNbO_3$ metasurface with a quasi-bound state at $\omega_1$ and a symmetry-protected mode near $2\omega_1$ in one meta-atom. (c) Periodically poled microring with pump and second-harmonic modes. (d) Real (solid) and imaginary (dashed) parts of $\chi^{(3)}_{\mathrm{casc}}$ versus $\Delta\omega$ in units of the second-harmonic linewidth $\gamma_2$, normalised to $|\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}|$ at $\Delta\omega = 2\gamma_2$.

## 2. Cascaded susceptibility at a second-harmonic resonance

A pump at frequency $\omega_1$ excites a mode $\mathbf{f}_1$ of the structure, with resonance frequency $\omega_a$ and energy decay rate $\gamma_1 = \omega_a/Q_1$, and its second harmonic falls near a mode $\mathbf{f}_2$ with frequency $\omega_2$ and decay rate $\gamma_2 = \omega_2/Q_2 \simeq 2\omega_1/Q_2$. Real fields are written as $\mathbf{E}e^{-i\omega t}$ + c.c., so that with $\mathbf{E}_1$ and $\mathbf{E}_2$ the complex amplitudes at $\omega_1$ and $2\omega_1$ the second-harmonic, difference-frequency and self-phase-modulation polarisations are $\varepsilon_0\chi^{(2)}\mathbf{E}_1\mathbf{E}_1$, $2\varepsilon_0\chi^{(2)}\mathbf{E}_2\mathbf{E}_1^*$ and $3\varepsilon_0\chi^{(3)}|\mathbf{E}_1|^2\mathbf{E}_1$ [31]. Under Kleinman symmetry one tensor $\chi^{(2)}_{ijk}(\mathbf{r})$ describes both second-order processes. For the field $a_1\mathbf{f}_1e^{-i\omega_1 t} + a_2\mathbf{f}_2e^{-2i\omega_1 t}$ + c.c., mode $n$ stores the energy $2N_n|a_n|^2$, with $N_n = \frac{1}{2}\int(\varepsilon_0\tilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2)\,d^3r$, $\mathbf{h}_n$ the magnetic profile and $\tilde{\varepsilon}_n = \partial(\omega\varepsilon_r)/\partial\omega$ at the resonance frequency of mode $n$. Projecting Maxwell's equations onto the two modes for slowly varying amplitudes [32, 33] gives

$$\dot{a}_1 = \left(i\delta_1 - \tfrac{\gamma_1}{2}\right)a_1 + i\beta_1^* a_1^* a_2 + i\alpha_3|a_1|^2 a_1 + \sqrt{\gamma_1^{\mathrm{e}}/2N_1}\,s, \tag{1}$$

$$\dot{a}_2 = \left(i\Delta\omega - \tfrac{\gamma_2}{2}\right)a_2 + i\beta a_1^2, \tag{2}$$

with the pump detuning $\delta_1 = \omega_1 - \omega_a$, the second-harmonic detuning $\Delta\omega = 2\omega_1 - \omega_2$, and $\gamma_1^{\mathrm{e}}$ the decay rate into the channel of an incident wave of power $|s|^2$. The couplings are $\beta = \varepsilon_0\omega_1 M/N_2$ and $\beta_1 = \varepsilon_0\omega_1 M/N_1$, with the overlap $M = \int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k}\, d^3r$, and the native Kerr coefficient is $\alpha_3 = (3\varepsilon_0\omega_1/2N_1)\int \chi^{(3)}_{ijkl} f^*_{1,i} f_{1,j} f_{1,k} f^*_{1,l}\, d^3r$. The factor 2 of difference-frequency generation makes $N_1\beta_1 = N_2\beta$, so that the energy $2(N_1|a_1|^2 + N_2|a_2|^2)$ is conserved in the absence of loss and drive (Supplement 1, Section 1).

When the pump envelope varies slowly on the time scale $|\Delta\omega + i\gamma_2/2|^{-1}$, Eq. (2) gives $a_2 = -\beta a_1^2/(\Delta\omega + i\gamma_2/2)$. Inserted into Eq. (1), this adds the term $i\alpha_{\mathrm{casc}}|a_1|^2 a_1$ with $\alpha_{\mathrm{casc}} = -\beta_1^*\beta/(\Delta\omega + i\gamma_2/2)$, which has the form of the native Kerr term. Writing $\alpha_3 = (3\varepsilon_0\omega_1/2N_1)\chi^{(3)}_{\mathrm{nat}}\int_{\mathrm{NL}} |\mathbf{f}_1|^4 d^3r$ defines the native susceptibility $\chi^{(3)}_{\mathrm{nat}}$ of the second-order material, where $\int_{\mathrm{NL}}$ extends over that material, and the analogous relation between $\alpha_{\mathrm{casc}}$ and $\chi^{(3)}_{\mathrm{casc}}$ defines the cascaded susceptibility,

$$\chi^{(3)}_{\mathrm{casc}} = -\frac{2\varepsilon_0\omega_1}{3N_2}\,\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int_{\mathrm{NL}} |\mathbf{f}_1|^4 d^3r} = -\frac{2\omega_1|\chi^{(2)}|^2\eta_{\mathrm{geom}}}{3n_2^2\,(\Delta\omega + i\gamma_2/2)}, \tag{3}$$

where $n_2$ is the smallest principal refractive index of the second-order material at $2\omega_1$, $|\chi^{(2)}|$ is the largest value of $|\chi^{(2)}_{ijk} u_i^* v_j v_k|$ over unit vectors ($2d_{33}$ for $LiNbO_3$; Supplement 1, Section 1), and

$$\eta_{\mathrm{geom}} = \frac{\varepsilon_0 n_2^2\,|M|^2}{|\chi^{(2)}|^2 N_2 \int_{\mathrm{NL}} |\mathbf{f}_1|^4 d^3r} \le 1 \tag{4}$$

is a geometric overlap that does not depend on the normalisation of the modes. The bound follows from the Cauchy–Schwarz inequality for modes with normal dispersion (Supplement 1, Section 1). The real part of $\chi^{(3)}_{\mathrm{casc}}$ is odd in $\Delta\omega$; for $\Delta\omega > 0$ it is negative and shifts the pump resonance to higher frequency [Fig. 1(d)]. The imaginary part is even in $\Delta\omega$ and positive, and describes an intensity-dependent loss of pump power into the second-harmonic mode, the cascaded analogue of two-photon absorption [14].

Equation (3) also follows without a coupled-mode equation for the second harmonic. Its field is the Green's function of the structure at $2\omega_1$ integrated against the source $\varepsilon_0\chi^{(2)}\mathbf{E}_1\mathbf{E}_1$, and inserting it into the difference-frequency polarisation gives a cascaded polarisation with the nonlocal kernel $\chi^{(2)}(\mathbf{r})G(\mathbf{r},\mathbf{r}';2\omega_1)\chi^{(2)}(\mathbf{r}')$. Near an isolated mode the pole term of $G$ [34,35] factorises the kernel into the overlaps $M$ and $M^*$, and projection onto $\mathbf{f}_1$ returns Eq. (3) (Supplement 1, Section 2).

For a uniform medium with refractive indices $n_1$ and $n_2$ at $\omega_1$ and $2\omega_1$ (isotropic, or with fields along the axis of smallest index) filling a periodic box, the pump mode at wavevector $k_1$ couples only to the second-harmonic mode at $2k_1$. When dispersion within the second-harmonic band is neglected, its frequency is $\omega_2 = 2\omega_1 n_1/n_2$ and $\eta_{\mathrm{geom}} = 1$ for aligned polarisations. Equation (3) then becomes $-2|\chi^{(2)}|^2/[3(n_2^2 - n_1^2)]$ to first order in $n_2 - n_1$. This is the susceptibility carried by the bound second-harmonic wave of a bulk crystal, which for $\Delta k = 2\omega_1(n_2 - n_1)/c$ equals the bulk result $-2\omega_1|\chi^{(2)}|^2/(3cn_2\Delta k)$ [3,4] (Supplement 1, Section 3). A resonator thus replaces the material mismatch $c\Delta k/n_2$, $0.04\,\omega_1$ in $LiNbO_3$, by $\Delta\omega$, $10^{-3}\omega_1$ for $Q_2 = 4000$ and $\Delta\omega = 2\gamma_2$; $\eta_{\mathrm{geom}}$ is the fraction of the bound-wave coupling that the two modes retain.

Equation (3) assumes a single second-harmonic mode, a stable steady state and a slowly varying pump envelope. Equations (1) and (2) omit the native self- and cross-phase modulation that involve the second-harmonic mode. For comparable overlaps these stay small as long as the cascaded shift of the pump resonance is small compared with both $|\Delta\omega|R/4$ and $|\Delta\omega|(R/2)^{1/2}$, with $R$ of Eq. (6) (Supplement 1, Section 4). Other second-harmonic modes, cascaded optical rectification with the electro-optic effect [36] and the longitudinal part of the bound second harmonic [37] add a background that varies little with $\Delta\omega$ across the resonance.

## 3. Limits and host materials

The second-harmonic mode sets the response time of the cascade. A component of $a_1^2(t)$ at frequency $\Omega$ measured from $2\omega_1$ enters Eq. (1) with the cascaded susceptibility $-C/(\Delta\omega + \Omega + i\gamma_2/2)$, where $C = 2\omega_1|\chi^{(2)}|^2\eta_{\mathrm{geom}}/(3n_2^2)$ (Supplement 1, Section 4). This response has a single pole at $\Omega = -\Delta\omega - i\gamma_2/2$ and is quasi-static for modulation frequencies small compared with $|\Delta\omega + i\gamma_2/2|$. At $\Delta\omega = 2\gamma_2$ it stays between 0.81 and 1.30 of its static value over a band of full width $\gamma_2/2\pi$, which is 0.1 THz for $Q_2 = 4000$ at a pump wavelength of 1.55 µm. The static value obeys $|\chi^{(3)}_{\mathrm{casc}}|\,|\Delta\omega + i\gamma_2/2| = C$ for any $Q_2$ and $\Delta\omega$, so raising $Q_2$ at fixed $\Delta\omega/\gamma_2$ increases $|\chi^{(3)}_{\mathrm{casc}}|$ and lowers its bandwidth $\Delta f$ in the same proportion. For any modulation band narrower than the distance of the pole from the origin, $\Delta f \le |\Delta\omega + i\gamma_2/2|/2\pi$, this identity gives

$$|\chi^{(3)}_{\mathrm{casc}}|\,\Delta f \le \frac{\omega_1|\chi^{(2)}|^2\eta_{\mathrm{geom}}}{3\pi n_2^2}, \tag{5}$$

whose right-hand side depends on neither $Q_2$ nor $\Delta\omega$ [Fig. 2(a)]. The pole also gives the sum rule $\int \mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}(\Omega)\,d\Omega = \pi C$ over the modulation frequency, so the integrated nonlinear absorption does not depend on the linewidth. Trade-offs between the strength and the bandwidth of resonantly enhanced nonlinearities have been discussed for slow-light and resonant nonlinear devices [38–40] and for harmonic generation [41], and appear in a cavity-enhanced cascaded frequency shifter [42].

Bulk $LiNbO_3$ at its natural phase mismatch has $|\chi^{(3)}_{\mathrm{casc}}| \approx 5\times10^{-21}$ m$^2$/V$^2$. In bulk crystals the cascaded response is nonlocal in time on a scale set by the phase and group-velocity mismatches [43], and for strong phase mismatch it is ultrafast and spans an octave [8], so its bandwidth exceeds 10 THz. A doubly resonant $LiNbO_3$ metasurface with $Q_2 = 4000$ and $\eta_{\mathrm{geom}} = 0.3$ reaches about $6\times10^{-20}$ m$^2$/V$^2$ over 0.1 THz. The product $|\chi^{(3)}_{\mathrm{casc}}|\,\Delta f$ of bulk cascading, above $5\times10^{-8}$ m$^2$ V$^{-2}$ Hz, therefore exceeds the bound of Eq. (5) for this metasurface, $1\times10^{-8}$ m$^2$ V$^{-2}$ Hz; the bound itself does not apply to the bulk response, which has no single pole. The resonator is preferable when the nonlinear element must be a sub-micrometre layer or must change sign over a small detuning.

The cascaded and native Kerr terms act on the same pump mode, so the field enhancement of the pump resonance cancels in their ratio $R = |\mathrm{Re}\,\alpha_{\mathrm{casc}}|/\alpha_3 = |\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}|/\chi^{(3)}_{\mathrm{nat}}$. At $\Delta\omega = 2\gamma_2$,

$$R \simeq \frac{Q_2\,\eta_{\mathrm{geom}}}{6}\,F, \qquad F = \frac{|\chi^{(2)}|^2}{n_2^2\chi^{(3)}_{\mathrm{nat}}}, \tag{6}$$

which neglects $(\gamma_2/2\Delta\omega)^2$ and overestimates $R$ by one part in 16. The host enters only through the dimensionless figure of merit $F$. With the constants of Supplement 1, Section 5, $F$ is 0.10 for $LiNbO_3$ and between $2.5\times10^{-3}$ and $4.5\times10^{-3}$ for GaP, AlN and $Al_{0.2}Ga_{0.8}As$, where the GaP value is an upper estimate. With $\eta_{\mathrm{geom}} = 0.3$, $R = 1$ is reached at $Q_2 \approx 200$ in $LiNbO_3$ and between 4000 and 8000 in the three semiconductors [Fig. 2(b)]. Estimates of the native Kerr index of $LiNbO_3$ at 1.55 µm span a factor of 1.8 and those of $d_{33}$ span 20%; together they place $R$ at $Q_2 = 4000$ between 11 and 30. GaAs is not included because its second harmonic at 775 nm lies above the band gap.

The imaginary part of Eq. (3) fixes the ratio of nonlinear loss to nonlinear refraction. It is $\rho = |\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}/\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}| = \gamma_2/(2|\Delta\omega|)$, which is 1/4 at $\Delta\omega = 2\gamma_2$. A Kerr resonator with this loss is bistable for some pump detuning only if $\rho < 1/\sqrt{3}$, and the same condition governs parametric gain of a pair of sidebands (Supplement 1, Section 4). When the cascaded term dominates, both therefore require $|\Delta\omega| > \sqrt{3}\gamma_2/2$; otherwise $\rho$ must be evaluated for the sum of the cascaded and native terms. Bistability also needs a pump detuning $|\delta_1|$ above $\sqrt{3}\gamma_1/2$ without nonlinear

loss and above $1.75\gamma_1$ at $\rho = 1/4$. Because $\Delta\omega$ is defined with the pump frequency, the cold resonator must then be built with the mismatch $2\omega_a - \omega_2 = \Delta\omega - 2\delta_1$.

The second-harmonic mode stores the energy of the pump mode multiplied by $S/|\Delta\omega|$, where $S = |\mathrm{Re}\,\alpha_{\mathrm{casc}}||a_1|^2$ is the cascaded shift of the pump resonance. In a continuous-wave steady state the elimination of $a_2$ is exact at any power. The steady state of Eqs. (1) and (2) can, however, lose stability through self-pulsing [11] once $S/|\Delta\omega|$ exceeds a value between about 1 and 10 that depends on $\gamma_1$ and $\delta_1$.

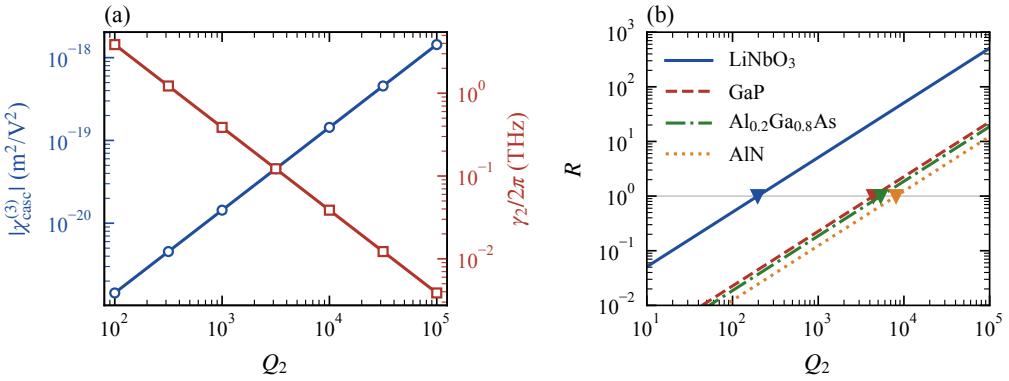



Fig. 2. (a) $|\chi^{(3)}_{\mathrm{casc}}|$ of $LiNbO_3$ at $\Delta\omega = 2\gamma_2$ (circles, left axis) and the bandwidth $\gamma_2/2\pi$ (squares, right axis) versus $Q_2$, for $\eta_{\mathrm{geom}} = 0.3$ and a pump wavelength of 1.55 µm. (b) $R$ from Eq. (6) versus $Q_2$ at $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$ for the hosts of Supplement 1, Section 5 (line styles); triangles and the horizontal line mark $R = 1$.

## 4. Metasurface implementation

### *4.1. Doubly resonant metasurface*

In a doubly resonant metasurface a quasi-bound state at $\omega_1$ couples the pump from free space [44], and a symmetry-protected bound state in the continuum near $2\omega_1$ holds the second harmonic [Figs. 1(b) and 3(a)]. Symmetry protects the latter only while the second harmonic lies below the first diffraction order of the substrate and the cladding [45, 46]. For a pump at 1.55 µm on silica this requires a period below 535 nm, at which a first-order pump resonance at the Γ point of the Brillouin zone would need an effective index above 2.9, out of reach in $LiNbO_3$. The pump mode then has to be folded to Γ by a perturbation that doubles the period, as in Ref. [47], and this perturbation opens weak second-harmonic diffraction channels; a suspended film raises the limit on the period to 775 nm. A doubly resonant cavity of this kind was proposed [48] and demonstrated [47] for second-harmonic generation. The overlap $M$ is nonzero only if the unit-cell symmetry of $\mathbf{f}_2$ matches that of $\chi^{(2)}$ contracted with $\mathbf{f}_1\mathbf{f}_1$, which sets selection rules for the $d_{33}$ orientation of a $LiNbO_3$ film [27]. At normal incidence both modes lie at Γ, which removes the bulk phase-matching condition. The mismatch $\Delta\omega$ can be tuned by temperature or by the electro-optic effect. Random variations of size and position lower the quality factor of the second-harmonic mode as the inverse square of their amplitude [49, 50], and an array of $N_p$ periods limits it to a value that grows as $N_p^2$ for a one-dimensional array [51]. The array and the pump spot must therefore span many periods. Measured quality factors of such modes are about $10^4$ in $Si_3N_4$ slabs near 740 nm [52], 800 for the second-harmonic mode of a GaN cavity at 775 nm [47] and above 8000 in $LiNbO_3$-on-insulator metasurfaces at 1.55 µm [53]; we have found no report of $Q_2 = 4000$ at 775 nm in $LiNbO_3$.

For $Q_2 = 4000$, $\eta_{\mathrm{geom}} = 0.3$ and $|\Delta\omega| = 2\gamma_2$, Eq. (6) gives $R \approx 20$ in $LiNbO_3$ [Fig. 3(b)]. For

$\Delta\omega < 0$ the cascaded and native terms add: self-phase modulation scales as $1 + R$, and four-wave mixing between sidebands within $|\Delta\omega + i\gamma_2/2|$ of the pump scales as $(1 + R)^2$ [Fig. 3(d)]. For $\Delta\omega > 0$ they subtract, and the net Kerr response reverses sign when $R > 1$. The pump energy needed for a given shift scales as $1/(1 + R)$; the second harmonic stores a further fraction equal to the cascaded shift divided by $|\Delta\omega|$ (Section 3), which is small for a shift of one pump linewidth when $\gamma_1 \ll |\Delta\omega|$ and reaches 0.5 to 0.8 at the folds of Fig. 3(c), where $\gamma_1 = \gamma_2$. The bistability threshold falls by less, by a factor of about $(1 + R)/5$ for $R$ between 10 and 100 at $|\Delta\omega| = 2\gamma_2$, because the cascaded loss raises the pump detuning needed for bistability [Fig. 3(c); Supplement 1, Section 4]. Cascaded bistability has been observed with a phase-mismatched second harmonic in a monolithic MgO:$LiNbO_3$ resonator pumped by a continuous-wave laser [54], and few-photon bistability in doubly resonant $\chi^{(2)}$ nanocavities has been analysed [13].

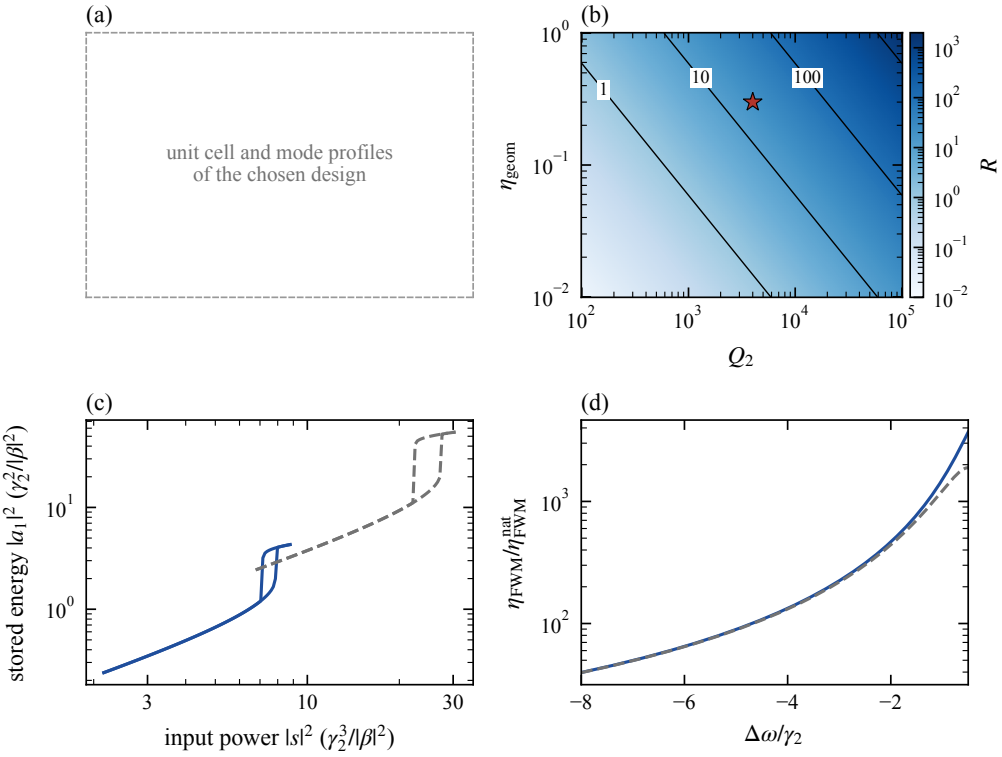



Fig. 3. Doubly resonant $LiNbO_3$ metasurface. (a) Unit cell with the pump and second-harmonic mode profiles. (b) $R$ from Eq. (6) versus $Q_2$ and $\eta_{\mathrm{geom}}$ at $\Delta\omega = 2\gamma_2$ (colour scale), with contours at $R = 1$, 10 and 100; the star marks $Q_2 = 4000$, $\eta_{\mathrm{geom}} = 0.3$. (c) Stored pump energy versus input power for increasing and decreasing sweeps of Eqs. (1) and (2), with (solid) and without (dashed) the second-harmonic mode, for $N_1 = N_2 = \frac{1}{2}$, $\Delta\omega = -2\gamma_2$, $\gamma_1 = \gamma_2$, $\gamma_1^{\mathrm{e}} = \gamma_1/2$, $\alpha_3$ chosen for $R = 20$, and in each case a pump detuning 1.3 times the minimum for bistability (a cold mismatch of $2.4\gamma_2$ in the cascaded case). (d) Ratio of four-wave-mixing efficiencies $\eta_{\mathrm{FWM}}$ with and without the second-harmonic mode at equal intracavity pump energy, $|\alpha_3 + \alpha_{\mathrm{casc}}|^2/\alpha_3^2$ (solid), and $(1 + R)^2$ with $R$ evaluated at each $\Delta\omega$ (dashed), versus $\Delta\omega/\gamma_2$ for $\Delta\omega < 0$ and the same $\alpha_3$.

### 4.2. Singly resonant metasurface and proposed measurement

Without a pump resonance the pump field in the meta-atom is fixed by the incident wave. With $\mathbf{e}_1$ the linear pump field per unit incident amplitude, the elimination of Section 2 applies with $\mathbf{f}_1$ replaced by $\mathbf{e}_1$. The overlap becomes $\widetilde{M} = \int \chi^{(2)}_{ijk} f^*_{2,i} e_{1,j} e_{1,k}\, d^3r$. For a meta-atom much thinner than the wavelength, where $\mathbf{e}_1$ has a uniform phase, $R$ takes the form of Eq. (6) with $\eta_{\mathrm{geom}}$ replaced by $\eta^{(1)}$, defined as in Eq. (4) with $\widetilde{M}$ and $\mathbf{e}_1$ in place of $M$ and $\mathbf{f}_1$ [Fig. 4(b); Supplement 1, Section 7]. The nonlinear phase per unit incident power loses the factor $Q_1^2$ of a pump resonance, and there is no bistability. The local pump field and the native response then change little while $\Delta\omega$ is scanned across the second-harmonic resonance, which makes this structure suited to measuring the cascaded susceptibility. *[Design: geometry, $Q_2$, $\widetilde{M}$, $\eta^{(1)}$, $R$ at $\Delta\omega = \pm 2\gamma_2$ and the single-pass phase; Fig. 4(a).]*

The part of the response odd in $\Delta\omega$ would be measured with a closed-aperture Z-scan [Fig. 4(c)] as a function of the pump wavelength, which moves $2\omega_1$ across $\omega_2$, in analogy with the angle-tuned measurement in KTP [3]. The cascaded phase stays within 8% of its continuous-wave value for pulses longer than $1.5/\gamma_2$ (intensity FWHM), about 2.5 ps for $Q_2 = 4000$ at 1.55 µm [Fig. 4(d)]. The beam must be wide enough that its angular spread stays within the angular bandwidth of the second-harmonic mode [55]. Near a Mie resonance the native response alone can change the sign of the effective nonlinear index [56]. That contribution follows the pump response, so the cascaded part is the component odd in $\Delta\omega$ when the pump transmission is flat across the scan. The second-harmonic conversion efficiency of the same sample, together with the radiative part of $\gamma_2$ and the collection efficiency, would fix $|\widetilde{M}|$.

## 5. Microring resonators

In a doubly resonant $\chi^{(2)}$ microring [Fig. 1(c)] the elimination of Section 2 reproduces the cavity expression of Refs. [16, 17], measured in AlN, with the coupling rate $g$ of the pump and second-harmonic modes in place of the overlap $M$ (Supplement 1, Section 8). The same rate sets the second-harmonic efficiency $\eta_{\mathrm{SHG}} = P_{\mathrm{SH}}/P_{\mathrm{in}}^2$ on double resonance, which is reported for such devices [59–61], and eliminating $g$ expresses the cascaded shift of the pump resonance through measured quantities. For a pump locked to its resonance and both modes critically coupled to the bus waveguide,

$$\frac{\delta\omega_{\mathrm{casc}}}{\kappa_1} = \eta_{\mathrm{SHG}} P_{\mathrm{in}} \frac{\kappa_2 \Delta\omega}{\Delta\omega^2 + \kappa_2^2} \tag{7}$$

when the nonlinear loss is neglected, where $\kappa_n = \gamma_n/2$ are the half-widths, $P_{\mathrm{in}}$ is the input power and a positive $\delta\omega_{\mathrm{casc}}$ raises the resonance frequency. Other couplings multiply the right-hand side by $(\kappa_1/2\kappa_1^{\mathrm{e}})(\kappa_2/2\kappa_2^{\mathrm{e}})$, where $\kappa_n^{\mathrm{e}}$ are the rates of coupling to the bus. Equation (7) contains no overlap, material constant or mode volume: the second-harmonic power measures the nonlinear loss, and the ratio $\rho$ of Section 3 converts it into the shift. For the periodically poled thin-film $LiNbO_3$ ring of Ref. [60], with $\eta_{\mathrm{SHG}} = 5 \times 10^6$%/W and a coupling factor of 1.6, it gives 0.09 half-widths, or 17 MHz, against 0.03 MHz for the native response, at 5 µW and $\Delta\omega = 2\gamma_2$ [Fig. 5(b)]. This power lies below the parametric threshold of the ring and below the power at which photorefraction lowers the efficiency [60]; the coupling rate fitted to the lineshape of the ring would give a shift 4.7 times larger. To our knowledge, no device has been characterised by $\eta_{\mathrm{SHG}}$ and the cascaded shift together; measuring the two on one ring would test Eq. (3) without a fitted parameter. Thermal and photorefractive shifts [62, 63] are comparable with or larger than the cascaded shift at these powers, but even in $\Delta\omega$ to leading order. An electro-optic dither of $\Delta\omega$ with lock-in detection would separate the odd cascaded part (Supplement 1, Section 8), as the Z-scan of Section 4.2 would in a metasurface.

Relative to the native Kerr response of the same ring, $R$ lies between $6 \times 10^2$ and $2.8 \times 10^3$ for the device of Ref. [60], depending on whether $g$ is taken from $\eta_{\mathrm{SHG}}$ or from the lineshape, and

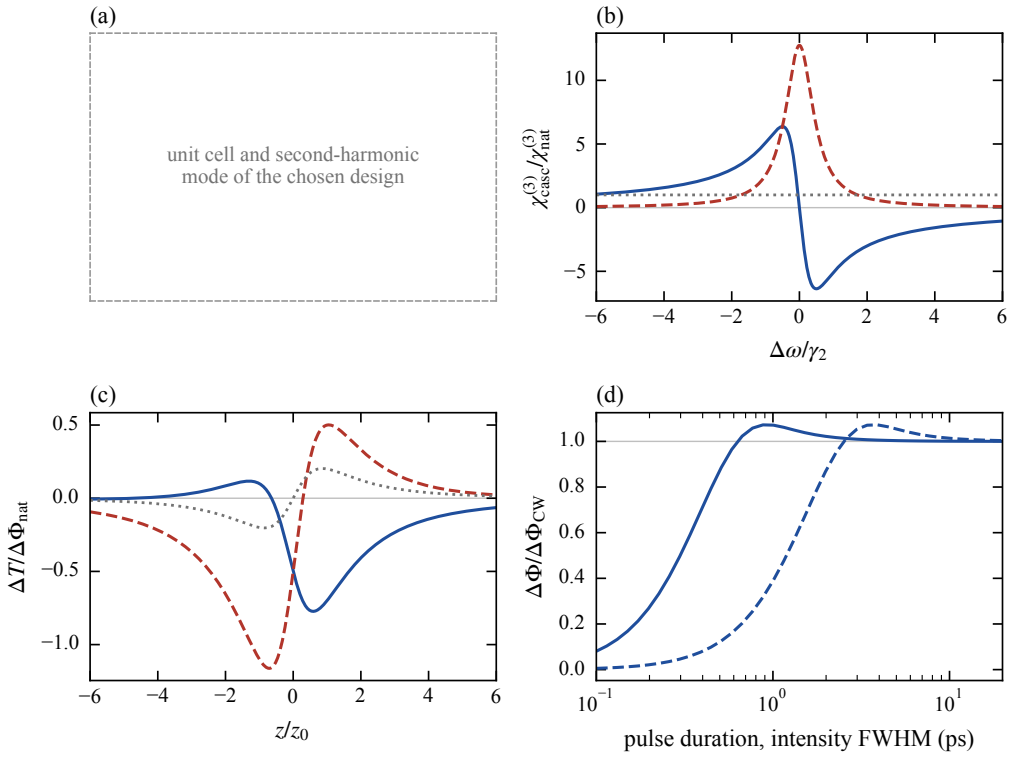



Fig. 4. Singly resonant $LiNbO_3$ metasurface. (a) Unit cell and second-harmonic mode profile. (b) Real (solid) and imaginary (dashed) parts of $\chi^{(3)}_{\mathrm{casc}}/\chi^{(3)}_{\mathrm{nat}}$ versus $\Delta\omega/\gamma_2$ for an illustrative value $R = 3$ at $|\Delta\omega| = 2\gamma_2$; the dotted line is the native response. (c) Closed-aperture Z-scan transmission change of a thin sample [57, 58], in units of the native phase $\Delta\Phi_{\mathrm{nat}}$, versus the sample position $z$ in units of the Rayleigh range $z_0$ ($z < 0$ before the focus), for $\Delta\omega = +2\gamma_2$ (solid), $\Delta\omega = -2\gamma_2$ (dashed) and the native response alone (dotted), including the nonlinear absorption of the cascade. (d) Intensity-squared-weighted Z-scan phase of the cascaded response for Gaussian pulses, relative to its continuous-wave value, versus the intensity FWHM of the pulse at $\Delta\omega = 2\gamma_2$, for $Q_2 = 10^3$ (solid) and $4 \times 10^3$ (dashed), at a pump wavelength of 1.55 µm. Horizontal lines mark 0 and 1.

between $4 \times 10^2$ and $1.8 \times 10^3$ for a ring with $Q_1 = 10^6$ and $Q_2 = 3 \times 10^5$. Without poling it is about 13 [Fig. 5(a)]. In the latter ring the parametric threshold falls from 9 mW for the native response to 8 µW, below the bistability threshold of 37 µW (Supplement 1, Section 8). Because the sign of the Kerr term follows that of $\Delta\omega$, one ring can provide either sign for modulation instability and comb formation [64–66].

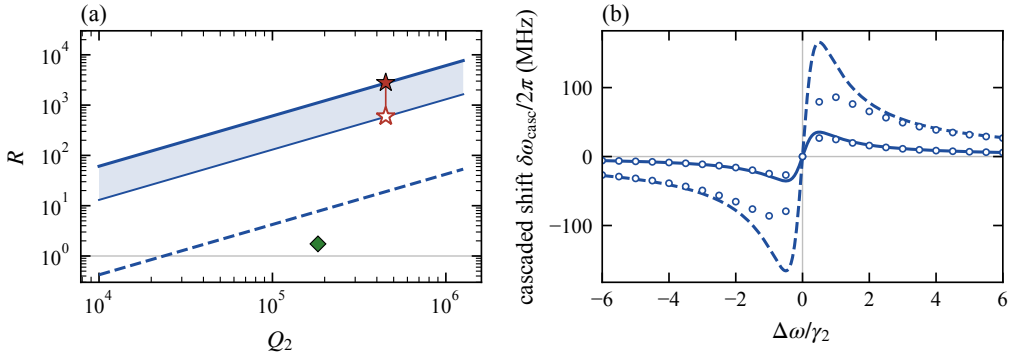



Fig. 5. Microrings. (a) $R$ versus $Q_2$ at a cold mismatch of $2\gamma_2$ for a periodically poled $LiNbO_3$ ring (design B of Supplement 1, Table S2) with the coupling rate scaled from the fitted value of Ref. [60] (upper solid line) or from its measured efficiency (lower solid line), and without poling (design C, dashed), with intrinsic $Q_2$ twice the loaded value. Stars: the device of Ref. [60] for the two coupling rates. Diamond: the AlN ring of Ref. [17], from its measured tuning of the self-phase modulation scaled to $\Delta\omega = 2\gamma_2$. The horizontal line marks $R = 1$. (b) Cascaded shift of the pump resonance of the device of Ref. [60] versus $\Delta\omega$ at 5 µW input, from Eq. (7) with the coupling factor of the device and the measured efficiency (solid) and with the efficiency implied by the fitted coupling rate (dashed). Circles: steady state with the pump locked to the shifted resonance, including the nonlinear loss. The pump half-width is $\kappa_1/2\pi = 178$ MHz.

## 6. Discussion and conclusion

Table 1. **Cascaded Kerr Response on Five Platforms** ($R$ **at** $\Delta\omega = 2\gamma_2$)

| Platform | Set by | $Q_2$ | $R$ | Bandwidth | CW threshold |
|---|---|---|---|---|---|
| Bulk $LiNbO_3$ | $\Delta k$ | – | $\approx 2$ | $> 10$ THz | bistability |
| Doubly resonant metasurface | $\Delta\omega$ | $4 \times 10^3$ | $\approx 20$ | $\approx 0.1$ THz | bistability |
| Singly resonant metasurface | $\Delta\omega$ | design | design | design | none |
| AlN microring [17] | $\Delta\omega$ | $1.8 \times 10^5$ | 1.7 [a] | $\approx 0.3$ GHz | – |
| Poled $LiNbO_3$ microring | $\Delta\omega$ | $3 \times 10^5$ | $4 \times 10^2$–$1.8 \times 10^3$ | $\approx 0.2$ GHz | 8 µW, parametric |

Bandwidth: response bandwidth for the bulk crystal and the linewidth of the narrower resonance otherwise, which is the second-harmonic mode for the metasurface and the pump mode for the rings. $R$ assumes $\eta_{geom} = 0.3$ for the doubly resonant metasurface and spans the two values of the coupling rate for the microring. CW threshold: the continuous-wave instability that appears first, observed for bulk $LiNbO_3$ in a cavity resonant at the pump [54] and predicted for the metasurface. For the microring the entry gives the input power at the parametric threshold; its bistability threshold is 37 µW. [a] From the measured tuning of the self-phase modulation, −2.7 to +4.7 times its calculated native value at the extrema of the tuning curve, $|\Delta\omega| = \gamma_2/2$ [17], scaled to $\Delta\omega = 2\gamma_2$.

A detuned second-harmonic resonance gives a cascaded Kerr nonlinearity of either sign, whose strength is set by $Q_2$, $\Delta\omega$ and $\eta_{geom}$ and which depends on the host only through $F$. In

$LiNbO_3$ we estimate it at about 20 times the native response in a doubly resonant metasurface and several hundred to a few thousand times in a poled microring (Table 1). Because Eq. (5) bounds the product of strength and bandwidth, a larger $R$ comes with a narrower bandwidth. The second-harmonic linewidth, which sets the response time of the cascade, is about 1 GHz in the ring and 0.1 THz in the metasurface, and bulk cascading, which is weaker, extends beyond 10 THz. Of the three, the metasurface is the only sub-micrometre layer addressed from free space, and its response is fast enough for picosecond pulses. Engineered second-order media would raise $R$ through $|\chi^{(2)}|^2$, as in coupled quantum wells with an interband $\chi^{(2)}$ above $10^3$ pm/V [67] and quantum-well metasurfaces [68], provided the native response and the absorption near the band edge [69], which also limit the second harmonic, stay small.

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**Data availability.** Write here something.

**Supplemental document.** See Supplement for supporting content.

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